Supporting Information: From Molecular Redox Motifs to Polymer Grafts: Attachment Thermodynamics and Neighboring-Site Effects

Working draft: authorship and affiliations to be confirmed

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Internal working draft. This SI contains finite method controls and implementation checks. Pending scientific analyses remain explicit.

1 Benchmark intervals and post hoc reference diagnostics

The complete panel has 39 endpoints from 30 molecules and seven studies. Molecule-cluster 95% percentile bootstrap intervals are [0.213,0.286] V for MAE, [0.245,0.313] V for RMSE and [0.128,0.257] V for bias. Study-cluster intervals are [0.178,0.306], [0.211,0.325] and [0.053,0.285] V, respectively. Both use 2,000 replicates and seed 20261021. The seven study clusters have unequal sizes; their percentile intervals are weakly calibrated. Molecule resampling retains common study and family effects and is also not a prediction interval for a new molecule.

Table S1 inventories all seven studies and the 39 selected endpoints. Each endpoint retains its exact molecular identity, redox-state pair, experimental value and source location in the accompanying data. Source-reported formal or midpoint potentials are retained under their stated measurement conditions; supporting-electrolyte and activity conventions are not assumed identical across studies. Values reported against another reference are used only with the source's own calibration.

Table S1: Primary experimental sources for the fixed MeCN benchmark. Counts are endpoints, not independent molecules. The table inventories the unchanged sourced panel; it adds no new measurements.

Study	Endpoints	Source location and reference handling
Attanayake et al.[1]	5	Table 1 and Figure 2; potentials reported versus Fc/Fc$^+$ in MeCN.
Cabral et al.[2]	2	Table 3; SCE-referenced values use the authors' measured internal Fc calibration, $E_{1/2}(\mathrm{Fc}) = 0.40$ V versus SCE.
Huynh et al.[3]	13	SI Table S3; source formal one-electron potentials versus Fc/Fc$^+$.
Wang et al.[4]	7	SI Table S1; TBAPF$_6$/MeCN columns and the stated reference convention.
Gerken and Stahl[5]	1	Results text and Figure 2; TEMPO midpoint reported directly versus Fc/Fc$^+$.
Daub et al.[6]	9	SI Table S1 and methods on page S10; imide midpoint potentials versus Fc/Fc$^+$.
Kim et al.[7]	2	Results text and Figure 1 discussion; exact source-parent measurements kept separate from bead properties.

For each cross-validation fold, a constant residual offset is estimated only from training endpoints and subtracted from the held-out prediction. Molecule and study folds remove all endpoints of the held-out molecule or DOI. The two-offset model fits separate constants for first and second redox steps. These choices were motivated by inspection of the complete retrospective panel and are internal diagnostics, not an untouched external test. Relative potentials to TEMPO, N-ethylphenothiazine, N-methylphthalimide and 1,4-benzoquinone cancel the computed ferrocene term algebraically. Each diagnostic excludes every endpoint of its anchor molecule. The resulting error includes the organic anchor's own model and measurement errors. Agreement after reference cancellation therefore does not uniquely identify a ferrocene error.

Table S2: Potential errors (V); primary unfitted statistics and internal offset diagnostics.

Holdout	Offset model	MAE	RMSE
None	Unfitted, primary	0.247	0.279
Endpoint	One constant	0.157	0.210
Molecule	One constant	0.158	0.211
Study	One constant	0.182	0.232
Endpoint	Separate step constants	0.106	0.166
Molecule	Separate step constants	0.106	0.166
Study	Separate step constants	0.124	0.177

Table S3: Fc-independent relative-potential diagnostics; all endpoints of each organic anchor are excluded. Errors include the anchor's own model and experimental uncertainty.

Organic anchor	Endpoints	MAE (V)	RMSE (V)
TEMPO	38	0.152	0.217
N-Ethylphenothiazine	38	0.152	0.219
N-Methylphthalimide	38	0.155	0.208
1,4-Benzoquinone	37	0.168	0.250

The spin-only partition includes multiplicity but omits an additional ferrocenium orbital contribution. Matrix spectroscopy and multireference calculations support a degenerate electronic ground state.[8] An idealized additional twofold factor at 298.15 K would lower ferrocenium G by 0.0178 eV, lower the computed Fc absolute potential by 0.0178 V and increase target potentials versus Fc by 0.0178 V. This limiting calculation would worsen the positive first-step bias. It is not an applied correction: solvent, orbital, vibronic and Jahn–Teller splittings are unresolved.

1.1 Fixed-geometry ferrocene functional controls

To test whether a change of reference functional supplies the hypothesized upward shift, we recalculated neutral and oxidized Fc single-point energies with three functionals at the same qualified R2SCAN-D4/def2-SVP/SMD geometries. All six calculations use def2-TZVPD, separately converged gas and MeCN SMD states, and the stated spin-contamination gate. The original functional reproduces the stored SMD energies within 10^{-6} eV. Nuclear thermochemistry is not recalculated, and no alternative absolute potential is assigned.

Table S4 separates gas electronic gaps from differential solvation. Relative to the primary functional, B3LYP and PBE0 lower the SMD electronic gap by 0.1184 and 0.0658 eV. A reference-only counterfactual subtracts each shift from the unchanged primary benchmark residuals. Its bias increases from +0.1896 V to +0.3080 and +0.2554 V, respectively. These functional-only changes therefore do not supply the proposed upward reference correction. They neither reproduce the full protocols of the published comparison values nor identify a unique source of the benchmark error; consistent alternative molecular and reference thermochemistry would be required for an alternative predictive protocol.

Table S4: Electronic-only Fc controls at fixed solution geometries. Gaps and solvation terms are in eV; counterfactual errors are in V. The latter hold every target-molecule and nuclear term fixed and are not alternative validated potentials.

Functional	Gas gap	Solvation difference	SMD gap	MAE	RMSE
ωB97M-V	6.4365	−2.0004	4.4361	0.2470	0.2793
B3LYP	6.3440	−2.0264	4.3177	0.3373	0.3700
PBE0	6.3525	−1.9822	4.3703	0.2947	0.3275

1.2 Nuclear rotor-cutoff sensitivity

The archived projected nuclear partitions at 25, 50 and 100 cm^{-1} were propagated through the same state and reaction sums with geometries, electronic energies, Hessians and spin factors fixed. No new SCF calculation was performed. The primary benchmark remains the 50 cm^{-1} result. Across all 39 endpoints, the largest individual span is 25.43 mV for the first duroquinone reduction, while the Fc reference spans 2.82 mV. Table S5 reports the aggregate metrics. These deterministic settings scenarios do not supply candidate-specific prediction intervals.

Table S5: Unfitted benchmark metrics under the three nuclear rotor cutoffs. Potentials and errors are in V; all 39 endpoints are retained.

Cutoff / cm^{-1}	MAE	RMSE	Bias
25	0.2449	0.2759	+0.1850
50 (primary)	0.2470	0.2793	+0.1896
100	0.2473	0.2800	+0.1901

For the 67 completed HCl-reference and 20 base-assisted reactions, the largest cutoff span is 4.46 kJ mol^{-1} and no nominal reaction sign changes. At each fixed cutoff, the contribution of a specified base/byproduct model remains identical across its five candidates. Variant reaction values use the exact SI eV-to-molar-energy factor; original source-conversion arithmetic is retained without replacement. Figure S1 displays these ranges. Independent reconstruction from the full archived state sums checks 117 potential evaluations and 261 reaction sums. This analysis does not test missing conformers, ion speciation, electronic-structure accuracy or attachment selectivity.

2 Neutral-partner diffuse-basis sensitivity

Five existing benchmark neutral partners were reoptimized with r^{2}SCAN, D4, def2-SVPD and SMD MeCN, with the same active ωB97M-V/def2-TZVPD electronic-energy protocol and a separately checked SVPD Hessian. Each passed stationarity, spin, finite/symmetry, two-column/two-step

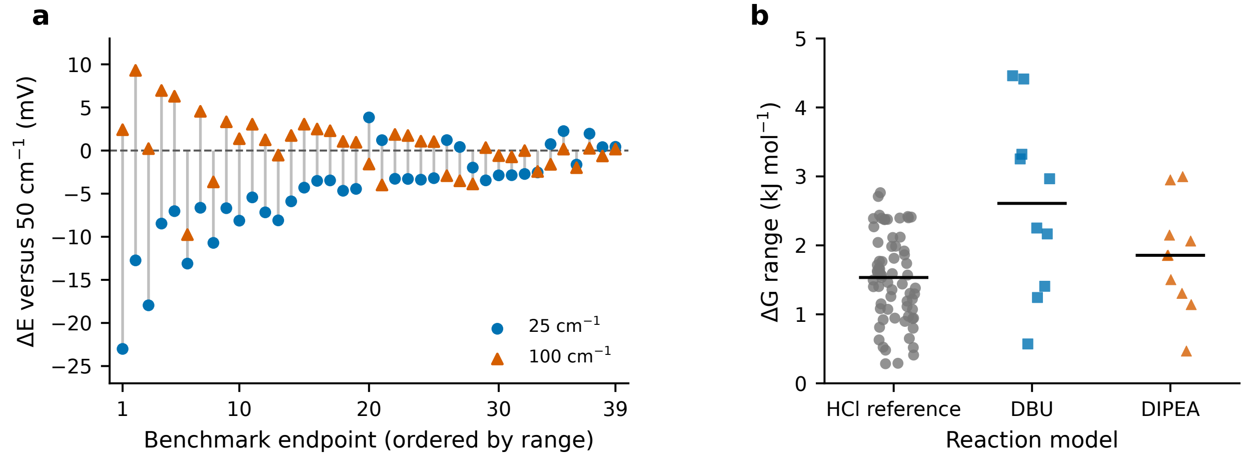


Figure S1: Nuclear rotor-cutoff sensitivity at fixed geometry and electronic structure. Panel a: potential shifts relative to 50 cm^{-1}, ordered by decreasing range over the three cutoffs; the dashed line denotes the primary convention. Panel b: 67 HCl-reference and ten DBU/ten DIPEA reaction-model ranges; horizontal marks are medians. Shared components make reaction values correlated. The ranges are settings scenarios, not confidence or candidate prediction intervals.

derivative and minimum checks. A basis-independent certified qRRHO integrator and an independent partition reconstruction were used. No primary state, protocol label or certificate was replaced. Table S6 separates the change in nuclear thermal contribution from the active-energy change due to reoptimization. The existing reduced state and Fc reference are held fixed; only the neutral-partner contribution to the first potential is tested. These are combined reoptimization/Hessian sensitivities; thermal changes are not isolated fixed-geometry basis effects, and conformational relaxation can contribute.

Table S6: SVPD minus primary neutral-partner contributions, in meV; the total also equals the first-potential shift in mV.

Neutral partner	Nuclear	Geometry/SP	Total
1,4-Benzoquinone	-11.29	-0.39	-11.67
1,2-Naphthoquinone	-13.06	-2.23	-15.29
9,10-Phenanthrenequinone	-7.78	-5.66	-13.44
N-Methylphthalimide	-8.63	-5.19	-13.82
N,N'-Diisopropyl-PMDI	-21.27	-13.81	-35.08

3 Exact selected cases: partial assessment with explicit holds

PMDI-ethyl and NDI-ethyl match enumerated exact designs by precursor, product and mapped new bond. All six redox states and the Fc reference were independently reconstructed; their existing grafting evidence hashes are unchanged. The registry now contains exact assessments rather than inherited family passes. Both routes remain held. The exact PMDI mono-N-ethyl/NH precursor is reported in EP0010440B1, Example 19 (printed page 10), with a stated NMR assignment.[9] That access report supports one requirement for PMDI; its selective bead attachment and experimental redox retention remain unverified. NDI precursor access is still unresolved in the checked collection. The soluble-polymer attachment precedent concerns phthalimide; separate molecular PMDI N-alkylation with CsF supports the reaction class but not the exact bead route.[10] Proposed DBU/MeCN conditions define the computed cycle and do not constitute a demonstrated synthesis protocol. Favorable model thermodynamics therefore do not override the route gate.

Table S7: Joined exact-case evidence. Potentials are V versus computed Fc; grafting free energies are kJ mol^{-1}. The finite PMDI occupancy cycle is complete; NDI occupancy thermochemistry remains missing.

Case	E_1	E_2	HCl route	DBU ions	Decision
PMDI-ethyl	-1.076	-1.977	-0.29	-103.07	Hold
NDI-ethyl	-0.764	-1.418	+10.05	-92.73	Hold

These records identify the missing evidence and avoid duplicate full-redox jobs, and link PMDI enumeration through molecular, route and finite occupancy calculations. They do not establish material eligibility or a converged polymer ensemble.

3.1 Source-supported TEMPO attachment and local oxidation model

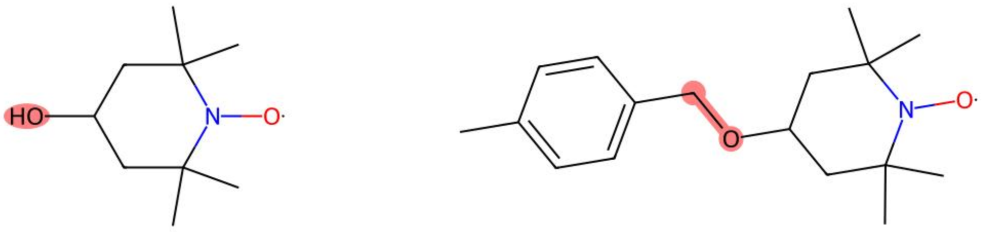
The 4-hydroxy-TEMPO precursor is attached through its hydroxyl oxygen to the benzyl site in the local model (Figure S2). The source reports this attachment to preformed Merrifield resin with NaH in DMF or THF and a washed-product N-oxyl content of 0.9 mmol g^{-1}.[11] The reported starting chloride density is referenced to the starting resin mass, so the two values do not directly determine conversion. The overall sodium-hydride stoichiometry is atom- and charge-balanced, with one nitroxide radical on each side; ion pairing and reaction free energy are not calculated.

For the exact local graft, the nitroxide/oxoammonium potential relative to the matched computed $\mathrm{Fc/Fc^+}$ reference is +0.576, +0.579 and +0.579 V for rotor cutoffs of 25, 50 and 100 cm^{-1}, respectively. The active solution electronic energies and four native thermal partitions were independently reconstructed. This is not an experimental potential for the graft, and the source reports catalytic use rather than battery cycling. Route evidence permits further evaluation; missing operating criteria, actual-condition grafting thermodynamics, intersite interactions and accessibility keep the material decision pending.

3.2 Directional electronic solution relaxation of the exact TEMPO graft

The oxidized O state has charge +1 and singlet multiplicity; the reduced R state has charge zero and doublet multiplicity. Each uses its previously checked r^{2}SCAN-D4/def2-SVP/SMD minimum. The shared heavy-atom ordering and full hydrogen connectivity match the declared graft. Their heavy-atom RMSD is 0.137 Å; the three bridge-torsion differences are 10.84, 2.24 and 1.88 degrees. The declared 0.3 Å and 30-degree diagnostics identify a paired local basin; they do not establish a conformer ensemble.

All six fresh calculations use ωB97M-V/def2-TZVPD with density fitting and native MeCN SMD. The native solvent constants are $\epsilon_s = 35.688$ and $\epsilon_{\mathrm{op}} = 1.80687364$. Optical polarization relaxes self-consistently during each vertical charge change while the initial slow polarization is frozen using a Pekar partition.[12] Each vertical event retains the same coordinates, basis and



4-Hydroxy-TEMPO precursor

4-Methylbenzyl local graft model

Figure S2: 4-Hydroxy-TEMPO and the capped local graft-site model. Highlighting identifies the precursor oxygen and the new oxygen–benzyl carbon bond. The drawing is a local molecular model, not a whole resin bead.

cavity for both electronic states. The native geometry-only CDS functional is included once; its treatment away from equilibrium is a stated model assumption, not a solvent-dynamics validation. With Q_X denoting the geometry of state X and D_X its equilibrium density,

$$L_{O\to R} = E_R^{\text{neq}}(Q_O; D_O) - E_R^{\text{eq}}(Q_R), \tag{1}$$

$$L_{R\to O} = E_O^{\text{neq}}(Q_R; D_R) - E_O^{\text{eq}}(Q_O). \tag{2}$$

The first direction separates into $E_R^{\text{eq}}(Q_O) - E_R^{\text{eq}}(Q_R)$ and $E_R^{\text{neq}}(Q_O; D_O) - E_R^{\text{eq}}(Q_O)$, with the analogous reverse expressions. The first component includes equilibrium solvation and geometry response; it is not an isolated gas inner-sphere term. No gas-inner or Born-outer quantity is added to either total. These electronic-structure energies include no nuclear thermal correction, explicit counterion or polymer environment.

Table S8: Checked electronic solution-relaxation components for the exact capped TEMPO graft, in eV. One paired minimum per charge state is used.

Direction	Equilibrium geometry	Frozen-slow increment	Total
Reduction $(+1 \to 0)$	0.396	1.081	1.477
Oxidation $(0 \to +1)$	0.553	1.086	1.639

The asymmetry diagnostic is $|L_{O\to R} - L_{R\to O}|/[(L_{O\to R} + L_{R\to O})/2]$. Its 5% cutoff is a predeclared diagnostic convention, not an experimental accuracy threshold. Independent reconstruction checks the six energy/density records, electron and spin traces, reference-density identities, and both directional sums. Both own-geometry energies reproduce the source active single points within 2×10^{-9} eV. The stationary minima are at the geometry level; active-single-point stationarity, global conformer convergence and experimental rates are unestablished. The exact-candidate ledger retains the two directions and leaves the symmetric scalar, actual-route grafting free energy, occupancy response and electronic coupling unfilled where their requirements have not passed.

3.3 Composition bounds and exact occupancy inputs

For neutral N–H/O–H substitution, the H-capped linear-fragment mass is

$$M = nM_{\text{C}_8\text{H}_8} + hM_{\text{CHCl}} + g(M_{\text{NuH}} - M_{\text{H}} - M_{\text{Cl}}) + 2M_{\text{H}} + M_{\text{extra}}, \tag{3}$$

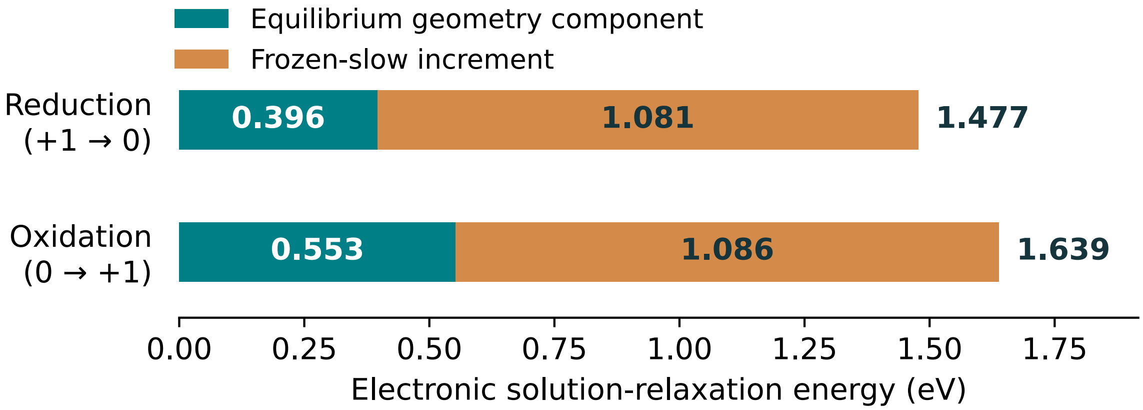


Figure S3: Directional electronic solution response of the same exact TEMPO graft used for the route and potential assessment. Components refer to the defined frozen-slow SMD model. The 10.4% asymmetry exceeds the declared 5% equal-curvature diagnostic; their average is not promoted to a qualified symmetric parameter. Statistical or prediction intervals are not estimated.

where n is repeat count, h is the number of original reactive handles and $g \leq h \leq n$ is graft count. The chloromethyl increment replaces a ring hydrogen; an ungrafted handle retains chlorine. Additional inactive mass M_{extra} is explicit and nonnegative. The capacity ceiling for an assumed z electrons per graft is $Q_{\text{upper}} = gzF/(3.6M)$. Actual utilization can be lower, including zero. This convention uses neutral dry-fragment mass and does not implicitly include crosslinker, absorbed electrolyte or charging counterions. Charged precursor/attachment classes require a separately declared counterion mass model.

All eight occupancy graphs at $n = 5, h = 3$ reproduce the analytic masses. Positional isomers with the same graft count have the same composition ceiling; their accessibility and interaction energies need not agree. The broader composition grid contains 21 (h, g) cases with one- and two-electron assumptions. No application target is fitted to these values, and the capacity dimension remains unresolved in the decision ledger.

Three exact PMDI inputs, occupied at sites 2, 3 or both 2 and 3, contain 115, 115 and 139 atoms, respectively. They preserve the open ten-carbon backbone, two plain-styrene padding units and any residual benzyl chlorides (Figure S5). Initial structures use seed 20261023, 32 distance-geometry embeddings and up to 2,000 MMFF steps; the best converged initial structure seeds each subsequent search. Connectivity, declared stereochemistry and geometry sanity are checked before calculation. Starting-structure and search records alone do not establish occupancy thermodynamics; each resulting state requires separate active-energy and native-Hessian qualification. The common empty reference has the same exact graph and configuration as the phthalimide empty control; no occupied phthalimide energy is substituted for a PMDI energy.

3.4 First-site PMDI grafting and the local-to-polymer comparison

The site-2 and site-3 polymer states pass native-Hessian admission and an independent reconstruction of the internal modes, nuclear partition and electronic contribution. The common empty state is the lowest-electronic-energy member of the two inherited empty-backbone conformers, selected by the declared baseline policy. All three retain the same backbone configuration, two styrene padding units and three original reactive-site positions. Each occupied graph contributes one sam-

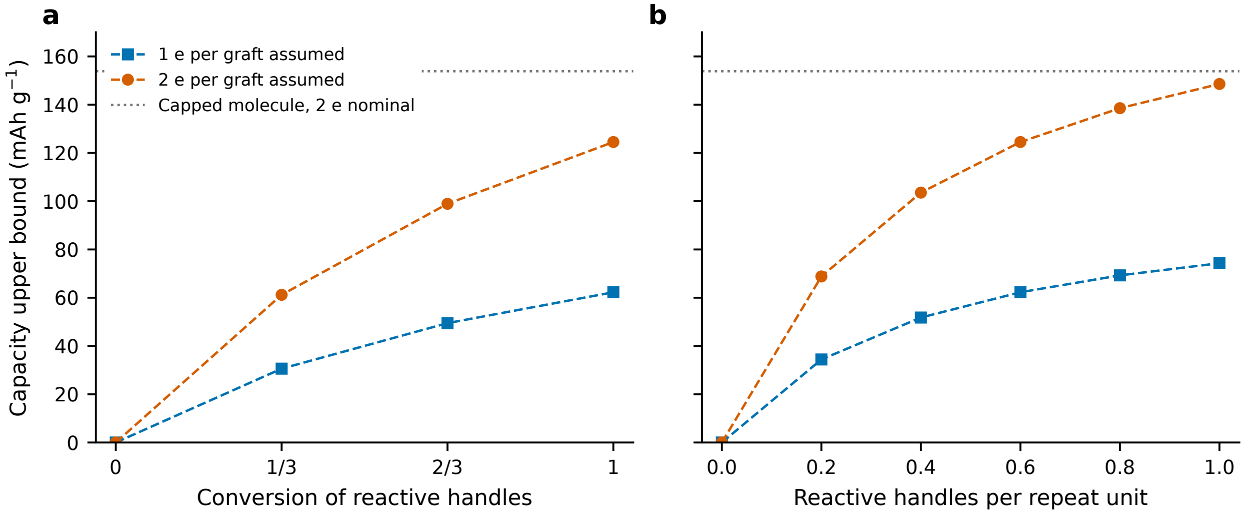


Figure S4: Composition-dependent PMDI capacity ceilings. The neutral H-capped fragment has five repeats; panel (a) fixes three reactive handles and varies conversion, while panel (b) varies handle density at full conversion. One- and two-electron budgets are assumptions. Residual chlorides and scaffold mass are retained. The dotted capped-molecule descriptor excludes the polymer scaffold; none of these curves establishes accessible material capacity.

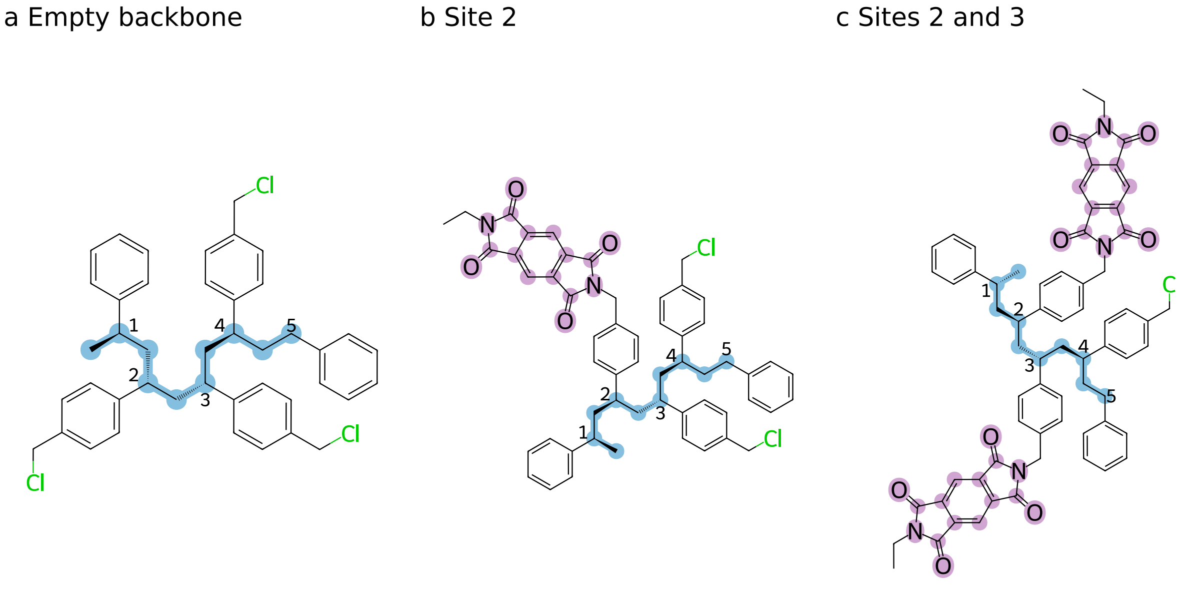

a Empty backbone
b Site 2
c Sites 2 and 3
Cl
Cl
Cl
1
2
3
4
5
O
N
O
O
O
N
Cl
Cl
1
2
3
4
5
O
N
O
O
O
N
Cl
1
2
3
4
5
O
N
O
O
O
N

Figure S5: Exact PMDI-ethyl occupancy inputs. Blue highlights the open backbone and purple the diimide cores. Panels show the empty chain, site 2 and sites 2/3; the separate site-3 reference is also prepared. Numbers label repeats. These two-dimensional drawings preserve connectivity and declared stereochemistry, not optimized spatial contact.

pled minimum; these values are finite structural scenarios rather than conformer-ensemble free energies.

For separated ions, the first-site reaction is $\mathrm{P_0 + HM + B \rightarrow P_i + BH^+ + Cl^-}$, where HM is the exact mono-N-ethyl/NH PMDI precursor and B is DBU. The contact-pair model replaces the two ionic products with one explicit $\mathrm{BH^+ \cdot Cl^-}$ species and includes the $\Delta n = -1$ standard-state term once. At 298.15 K this term is -7.9259 kJ mol^{-1}. The balances are checked independently for atoms and charge. The relative reaction $\mathrm{P_0 + RM \rightarrow P_i + RCl}$ gives the polymer-context penalty without the common precursor or base terms. The new reconstruction uses 96.48533212331001 kJ mol^{-1}/eV; earlier source arithmetic is retained.

Table S9: First-site PMDI grafting in sampled five-repeat models. All values are kJ mol^{-1} at 298.15 K in SMD MeCN. The local capped model has DBU/separated-ion $\Delta G = -103.07$ kJ mol^{-1} at the primary 50 cm^{-1} cutoff. The last column is the balanced polymer-minus-local difference; common reagent terms cancel.

Site	Cutoff / cm^{-1}	Separated ions	Contact pair	Polymer minus local
2	25	−97.03	−102.98	+6.61
2	50	−97.72	−102.52	+5.34
2	100	−97.15	−100.52	+6.11
3	25	−97.17	−103.12	+6.47
3	50	−96.96	−101.75	+6.11
3	100	−96.95	−100.32	+6.31

The site-2 electronic energy is 8.15 kJ mol^{-1} below site 3, while their primary free-energy difference is only -0.77 kJ mol^{-1}. The site-2-minus-site-3 free-energy differences at 25, 50 and 100 cm^{-1} are $+0.14$, -0.77 and -0.20 kJ mol^{-1}, respectively. The nominal placement preference therefore changes sign within this settings test. It cannot support positional selectivity or an experimental occupancy distribution. The paired-state calculation now passes the same native-Hessian and independent reconstruction gates. Its four-state electronic nonadditivity is $\Delta\Delta E = -2.31$ kJ mol^{-1}; the corresponding primary free-energy result is $\Delta\Delta G = -1.86$ kJ mol^{-1}. Neither quantity isolates motif binding. Broader conformer searches remain necessary for both the first-site and neighboring-site comparisons.

3.5 Completed finite PMDI occupancy cycle

The paired state has 411 internal modes, a lowest frequency of $+5.77$ cm^{-1} and a largest Cartesian gradient of 7.62×10^{-6} hartree bohr^{-1}. Its Hessian asymmetry is 1.54×10^{-10} atomic units, and the largest relative error over the two sampled columns at both displacements is 3.43×10^{-6}. Independent mode projection and nuclear equations reconstruct the accepted free energy. Table S10 specifies the matched states; the shared empty reference follows the stated lowest-electronic-energy policy rather than a claim of global conformer convergence.

At the primary setting, DBU/separated-ion attachment at site 2 is -97.72 kJ mol^{-1} on the empty chain and -99.59 kJ mol^{-1} when site 3 is grafted. The corresponding site-3 values are -96.96 and -98.82 kJ mol^{-1}. Both attachment orders give -196.54 kJ mol^{-1} for the direct double-grafting reaction. The contact-pair calculation gives the same -1.86 kJ mol^{-1} nonadditivity: the shared reaction constant and standard-state term cancel algebraically. Rotor cutoffs of 25/50/100 cm^{-1} give $-1.07/-1.86/-1.82$ kJ mol^{-1}. These are deterministic settings comparisons, not statistical uncertainty bounds.

Table S10: Matched neutral PMDI occupancy states at the primary 50 cm^{-1} rotor cutoff. Each graph contributes one native free energy; values are eV.

Occupied sites	Atoms	G
None	91	-82874.029197
2	115	-94077.698759
3	115	-94077.690815
2,3	139	-105281.379671

The accepted paired geometry has core centroids 16.30 Å apart and a 69.0° interplane angle (Figure S6). The finite result includes backbone, conformational and solvent response; it is not an estimate of intrinsic close-contact core binding or electronic coupling. No loading distribution, atactic ensemble or preferred experimental graft site is inferred.

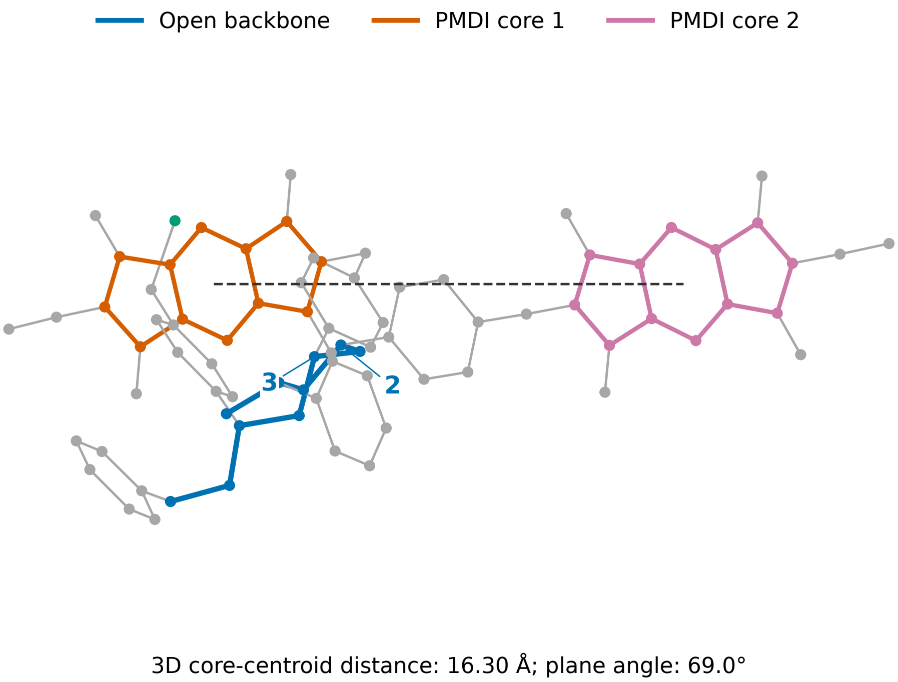


Figure S6: Actual paired PMDI minimum used in the matched cycle. A proper rotation makes both conjugated cores visible while preserving the saved XYZ coordinates. Blue highlights the open backbone; numbers mark repeats 2 and 3. Orange and magenta mark the PMDI cores, and green marks the remaining chloride at reactive site 4. Repeats 1 and 5 are plain-styrene padding. Hydrogens are omitted and bond lines follow the checked covalent graph. The dashed line joins core centroids; the stated distance is three-dimensional.

4 Relative-basicity controls

The balanced reactions $\mathrm{B} + \mathrm{DBUH}^{+} \rightarrow \mathrm{BH}^{+} + \mathrm{DBU}$ avoid HCl, chloride and a free-proton reference. The exact pyridine, triethylamine and DBU identities match the revised author-maintained

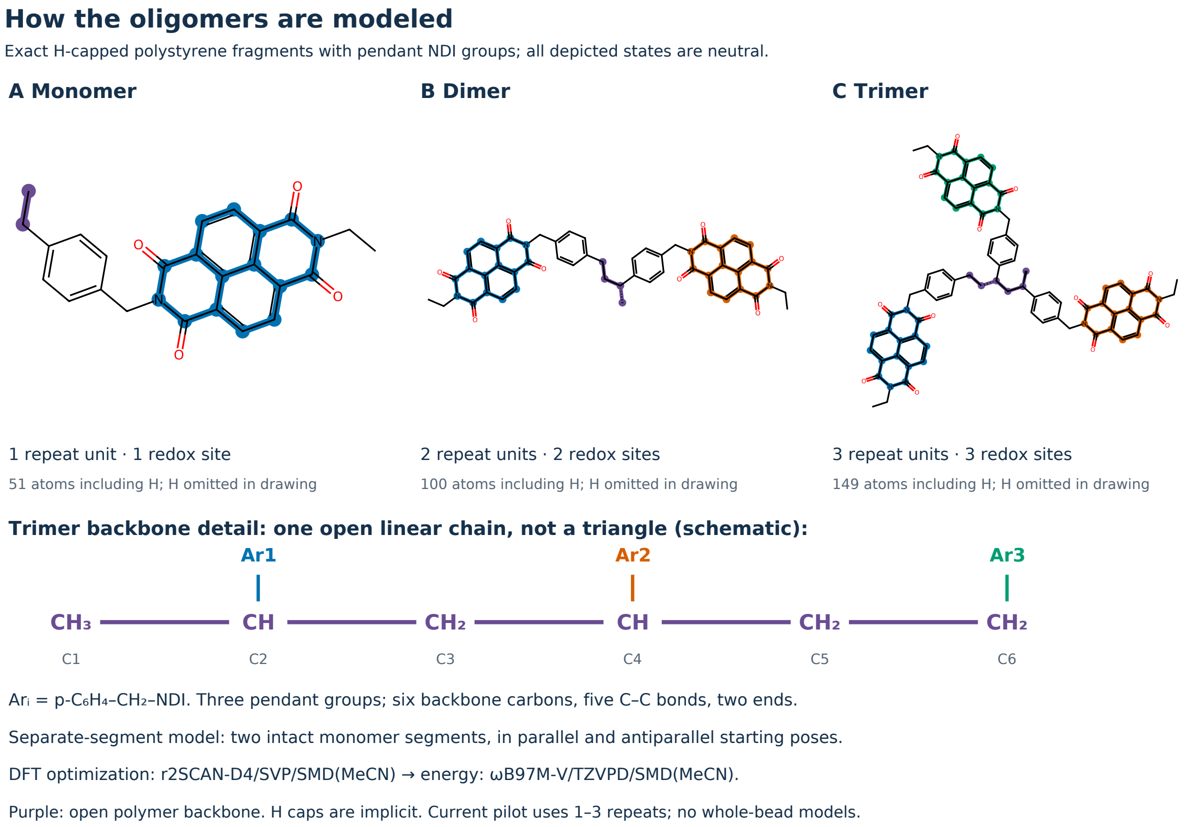
MeCN scale, with conjugate-acid pKa values 12.53, 18.83 and 24.31.[13, 14] Four new components and the two reused DBU components pass the same thermodynamic checks and an independent six-partition reconstruction. At 298.15 K, computed values are 71.82 and 32.68 $\mathrm{kJ\,mol^{-1}}$, compared with pKa-derived values 67.24 and 31.28 $\mathrm{kJ\,mol^{-1}}$; differences are +4.58 and +1.40 $\mathrm{kJ\,mol^{-1}}$. These values use the exact SI eV-to-molar-energy factor. Historical source records use 96.485 $\mathrm{kJ\,mol^{-1}}$/eV and are retained without replacement. Source temperature/activity qualifications remain open. This two-reaction control does not validate DIPEA, contact pairs or the complete grafting reaction and cannot assign the DBU/HCl discrepancy to a unique species. The separated model uses bare chloride in a continuum. Recent MeCN ion-solvation comparisons identify both species-dependent errors and sensitivity to absolute single-ion conventions, so the capture discrepancy cannot be assigned to chloride alone.[15] Acid–chloride homoconjugation, such as $\mathrm{HCl} + \mathrm{Cl}^- \rightleftharpoons \mathrm{HCl}_2^-$, would introduce an additional species and acid-dependent speciation. It is not included in either ionic representation; no population or homoconjugation constant is inferred from the free-ion pKa comparison. Spectrophotometric titration against bromocresol green reports $\log K^{\mathrm{H}} = 18.64(1)$ for DIPEA in MeCN at (25.0 ± 0.1) °C.[16] A separate NMR competition study estimates pKa 18.7 for diisopropylethylammonium hexafluorophosphate in MeCN-d_3.[17] The latter uses a 0.02 M initial salt solution and triethylamine reference; its supporting table labels negative values as a K_a ratio and applies an absolute value in the reported logarithm. We retain the authors' stated estimate rather than independently refitting its sign convention. The two sources are not pooled into an uncertainty interval. Combining the 18.64 value with pKaH(DBU)=24.31 and 298.15 K gives +32.36 $\mathrm{kJ\,mol^{-1}}$ for $\mathrm{DIPEA} + \mathrm{DBUH}^+ \rightarrow \mathrm{DIPEAH}^+ + \mathrm{DBU}$, compared with +31.80 $\mathrm{kJ\,mol^{-1}}$ from the stored components. Indicator calibration and activity conventions are not established to match the separate basicity scale. The 0.56 $\mathrm{kJ\,mol^{-1}}$ difference is not an uncertainty estimate or proof of an error in a particular species.

With pKa(HCl)=10.30, the 18.64 value instead implies −47.60 $\mathrm{kJ\,mol^{-1}}$ for the separated capture term, compared with −70.98 $\mathrm{kJ\,mol^{-1}}$ computed. Holding each computed HCl-reference reaction fixed and replacing only the capture term yields +10.03/+8.43 $\mathrm{kJ\,mol^{-1}}$ for methoxy/trimethyl BQ, compared with −13.34/ − 14.94 $\mathrm{kJ\,mol^{-1}}$ for the all-DFT DIPEA sums. This hybrid changes shared-error cancellation because HCl cancels from the complete all-DFT cycle; it is a sensitivity, not a calibrated reaction prediction or an accepted uncertainty interval. Salt speciation, chloride homoconjugation, attachment selectivity and actual grafting yields remain unvalidated. The earlier 18.6 value is a distinct PCM estimate despite a conflicting generic table label and is not treated as an independent measurement.[18]

5 Finite-polymer method controls

Twenty-six of the original twenty-seven electronic oligomer cases currently pass geometry and active-energy checks. The open PMDI trimer has recovered and passed independent force checks. The stacked-start PMDI trimer returned optimizer convergence and saved active energies, but a later GPU-library call terminated abnormally. Separate final-force checks disagree with finite energy differences and remain unresolved, including a subsequent finer-step diagnostic. This case stays outside completed comparisons; no minimum or free energy is inferred from optimizer convergence alone. Construction and XYZ audits verify the intended open-chain topology. A trimer contains six backbone carbons, five backbone C–C bonds and two chain ends, rather than a triangular or cyclic backbone (Figure S7). Proximity of pendant rings does not change that covalent connectivity.

For the six-graph spacing control, all twelve electronic conformers qualify. The selected electronic nonadditivities are −10.98 and −14.50 $\mathrm{kJ\,mol^{-1}}$ for adjacent and one-spacer grafts, respec-


How the oligomers are modeled
A Monomer
B Dimer
C Trimer
O
N
1 repeat unit · 1 redox site
2 repeat units · 2 redox sites
3 repeat units · 3 redox sites
Ar1
Ar2
Ar3
CH$_3$
CH
CH$_2$
CH
CH$_2$
CH$_2$
C1
C2
C3
C4
C5
C6

Exact H-capped polystyrene fragments with pendant NDI groups; all depicted states are neutral.

51 atoms including H; H omitted in drawing

100 atoms including H; H omitted in drawing

149 atoms including H; H omitted in drawing

Trimer backbone detail: one open linear chain, not a triangle (schematic):

Ar_i = p-C_6H_4–CH_2–NDI. Three pendant groups; six backbone carbons, five C–C bonds, two ends.

Separate-segment model: two intact monomer segments, in parallel and antiparallel starting poses.

DFT optimization: r2SCAN-D4/SVP/SMD(MeCN) → energy: ωB97M-V/TZVPD/SMD(MeCN).

Purple: open polymer backbone. H caps are implicit. Current pilot uses 1–3 repeats; no whole-bead models.

Figure S7: Exact finite-polymer construction used in the interaction controls. The highlighted open backbone distinguishes covalent connectivity from spatial contacts among pendant groups. Monomer, dimer and trimer models are dense local scenarios and do not represent an experimentally measured sequence or graft-loading distribution.

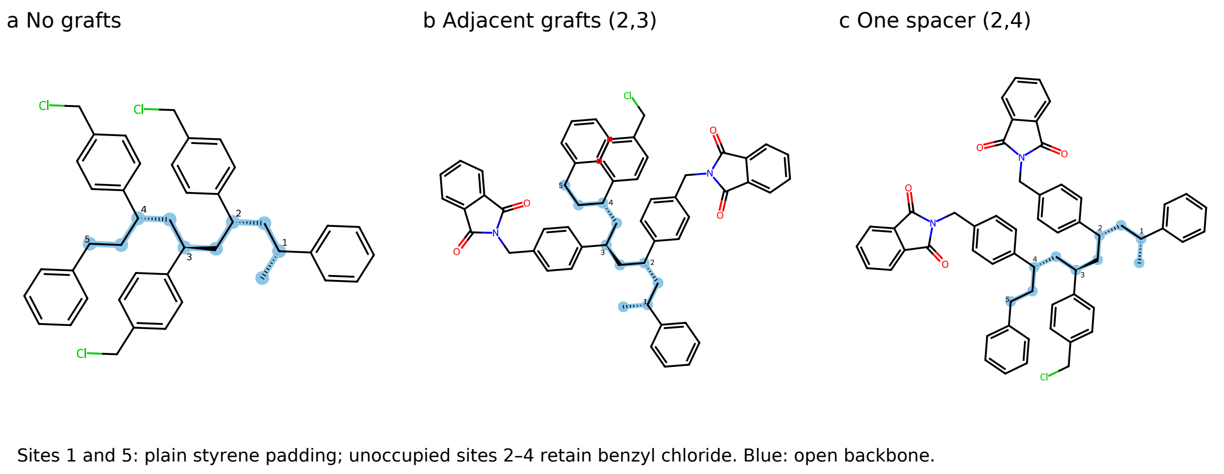
tively. These refer to a matched empty-backbone minimum within the declared, limited sampling set. It demonstrates why a matched empty-backbone reference is necessary; it does not establish a universal distance trend.

All twelve pose-specific native free energies now qualify and have been independently reconstructed from their electronic, Hessian and projected partition records. For a finite set of distinct conformers, we compute

$$G_{\text{finite}} = -RT \ln \sum_{c\in\mathcal{C}} \exp[-G_c/(RT)]. \tag{4}$$

Each distinct optimized minimum has unit occurrence weight; conformer-search hit counts are not degeneracies. Complete-link grouping checks mapped heavy-atom RMSD and electronic-energy agreement. RMSD cutoffs of 0.05, 0.10 and 0.20 Å leave these six pairs distinct, with pairwise RMSDs of 3.50–6.53 Å. The finite partition uses the same temperature, solvent and free-energy protocol for every graph. A partition over a sampled subset does not establish the complete equilibrium ensemble.

The spatial comparison in the main text uses both accepted conformers of each doubly occupied graph. Conjugated core-ring atoms are identified from the graft fragment and independently



a No grafts

b Adjacent grafts (2,3)

c One spacer (2,4)

Sites 1 and 5: plain styrene padding; unoccupied sites 2–4 retain benzyl chloride. Blue: open backbone.

Figure S8: Exact five-repeat phthalimide graphs used in the occupancy cycles. Blue highlights the open ten-carbon backbone. Sites 1 and 5 are plain styrene padding; unoccupied reactive sites retain benzyl chloride. Numbers label repeat positions, and stereobonds retain the declared configuration. These are connectivity drawings, not sampled spatial conformations. Single-occupancy reference graphs are included in the evidence.

mapped to the saved XYZ connectivity. Core centroids are the arithmetic means of those coordinates; core planes are fitted by singular-value decomposition and their acute interplane angle is reported. Orthographic views apply proper rotations with determinant +1 and a common spatial scale. Within a graph, the displayed conformer difference is $G_i - \min_j G_j$ over its two known conformers, using the primary 50 cm^{-1} cutoff. This quantity includes the entire chain and solvent response. It is separate from the four-state occupancy cycle. Coordinate data, view matrices and source identities accompany the figure; cached-coordinate rendering reproduces its pixels with original calculation paths blocked, without repeating physical qualification.

The primary adjacent and one-spacer finite-set $\Delta\Delta G$ values are -5.17 and -11.39 kJ mol^{-1}, respectively (Figure S9). Values at rotor cutoffs of 25/50/100 cm^{-1} are $-5.13/-5.17/-4.64$ and $-12.34/-11.39/-11.69$ kJ mol^{-1}. These are partition-model sensitivities, not confidence intervals or bounds on undiscovered-conformer error. Relative to the lowest-G member of each two-pose set, the largest configurational lowering is 0.140 kJ mol^{-1}. The previously reported values, $-5.15/-11.23$ kJ mol^{-1}, used only the minimum-electronic-energy geometry from each graph. Both sets remain tied to the same limited baseline sampling and single backbone configuration.

The largest Cartesian gradient component ranges from 6.15×10^{-6} to 7.62×10^{-5} hartree bohr^{-1} across these twelve conformers, below the 4.5×10^{-4} acceptance threshold. Table S11 reports the stored physical checks for every pose. These values describe the checked stationary structures and do not assess conformer-search completeness. The -2.72 cm^{-1} mode falls within the declared tolerance and enters qRRHO as its absolute value.

For DBU and separated ions, the finite-partition free energy of grafting onto an empty backbone is -110.86 kJ mol^{-1} at site 3 and -103.37 kJ mol^{-1} at site 4. With site 2 already grafted, the values become -116.02 and -114.76 kJ mol^{-1}. The contact-pair alternatives are -115.65 to -120.82 kJ mol^{-1} for site 3 and -108.17 to -119.56 kJ mol^{-1} for site 4. Both attachment orders close the same nonadditivity cycle. The base/byproduct contribution cancels from this conditional difference, as does the common empty-chain term when comparing the two spacing cycles. Con-

Table S11: Stored native physical checks for the twelve spacing-control conformers. Gradients are maximum Cartesian components in hartree bohr^{-1}; frequencies are the lowest projected internal modes in cm^{-1}; asymmetry is the maximum absolute Hessian asymmetry in atomic units. The final column is the largest relative-norm error among the sampled finite-difference Hessian columns, expressed as a percentage.

Occupied sites	Pose	Maximum gradient	Lowest frequency	Asymmetry	Column error (%)
empty	0	6.15×10^{-6}	10.60	3.68×10^{-8}	0.342
empty	1	1.01×10^{-5}	10.76	2.04×10^{-7}	0.005
2	0	3.53×10^{-5}	8.72	1.77×10^{-10}	0.009
2	1	4.22×10^{-5}	7.62	1.67×10^{-11}	0.005
3	0	5.50×10^{-5}	3.45	7.21×10^{-11}	0.008
3	1	8.39×10^{-6}	7.61	3.97×10^{-10}	0.006
4	0	2.47×10^{-5}	6.60	6.73×10^{-11}	0.007
4	1	8.81×10^{-6}	-2.72	3.98×10^{-10}	0.004
2,3	0	4.63×10^{-5}	10.04	1.35×10^{-10}	0.008
2,3	1	2.80×10^{-5}	6.36	6.73×10^{-9}	0.004
2,4	0	2.75×10^{-5}	7.50	9.21×10^{-12}	0.004
2,4	1	7.62×10^{-5}	7.23	2.05×10^{-8}	0.007

former response in the occupied graphs remains relevant. More favorable subsequent attachment is not automatically desirable when weak unwanted neighboring-site interaction is the objective.

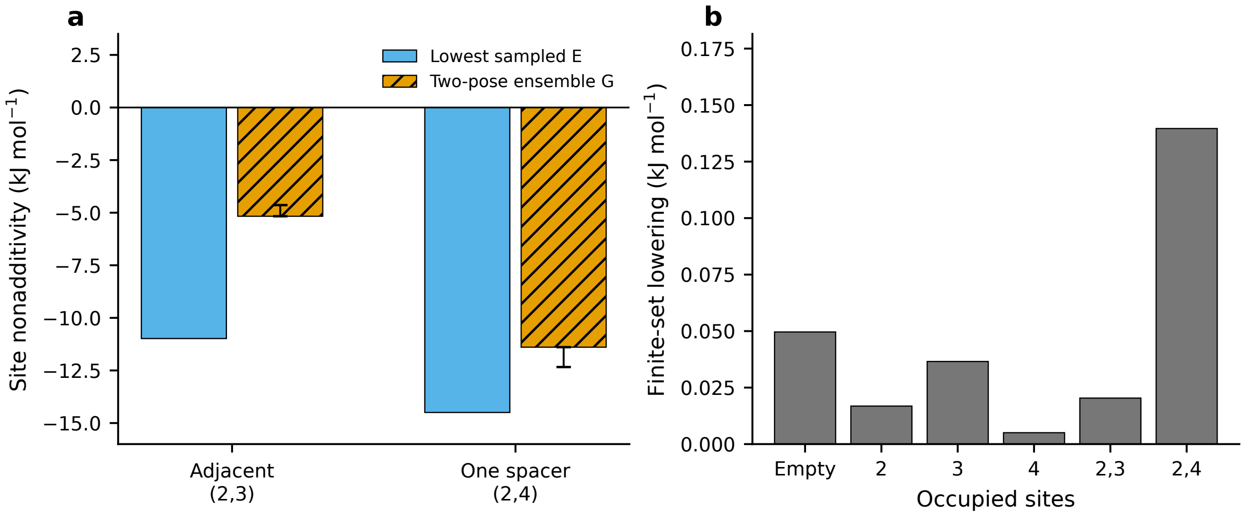


Figure S9: Finite-set phthalimide occupancy control in MeCN at 298.15 K. Panel (a) distinguishes lowest sampled electronic-energy cycles from two-pose free-energy partitions; whiskers vary only the rotor cutoff. Panel (b) shows the configurational lowering within each known two-pose set. Twelve native records pass their checks, but these two-pose sets do not establish converged conformer ensembles or a general spacing trend.

To examine local contacts, each phthalimide group and its benzyl carbon was retained at the parent geometry and capped with hydrogen where it had been connected to the polymer aryl group. For each of four parent poses and three cap lengths (1.07, 1.09 and 1.11 Å), independent acetonitrile-model electronic energies for the pair and both capped fragments give $E_{AB} - E_A - E_B$. The twelve triplets use the same frozen coordinates and electronic settings within each comparison. At 1.09 Å, the adjacent close/separated poses give -24.76 and $+0.048$ kJ mol^{-1}, and the one-spacer close/separated poses give -26.22 and $+0.061$ kJ mol^{-1}, respectively. The ranges over cap lengths are -24.81 to -24.71, $+0.047$ to $+0.049$, -26.24 to -26.20 and $+0.060$ to $+0.062$ kJ mol^{-1}, respectively. These uncorrected electronic values include the caps and continuum response; they are not ensemble free energies, electron-transfer couplings or decompositions of the whole-chain occupancy cycle. Basis-superposition error and broader conformer sampling remain unresolved.

The finite conformer sets do not establish convergence with respect to broader sampling or backbone configuration. Consequently, these conditional-cycle values cannot establish a general spacing design rule or the behavior of an exact redox motif. The final stacked-trimer structure is excluded because it failed the final force check.

5.1 Broader sampling and retained-state reconstruction

A separate smooth-surface acetonitrile model provides a numerically controlled setting for broader phthalimide sampling; its definition and small-molecule checks are described below. Twenty inputs were declared for each of twelve occupancy/backbone-configuration groups. Both empty-backbone configurations have checked native thermochemistry for all twenty inputs. The forty empty inputs comprise 38 diverse warm starts from saved CREST/GFN2-xTB searches and two retained DFT reference structures.[19] The site-2 sets contain 18 and 16 checked states for the all-same and alternating configurations, respectively. These configuration labels specify two backbone structures, rather than an experimental atactic-polymer distribution.

Full connectivity, hydrogen attachment and configured stereochemistry are checked before symmetry-aware heavy-atom alignment. Chirality-preserving graph mappings and proper rotations identify duplicates without admitting reflected structures. At RMSD thresholds of 0.05, 0.10 and 0.20 Å combined with a 0.20 kJ mol^{-1} electronic-energy tolerance, the all-same and alternating empty sets contain 20 and 19 distinct structures, respectively. Alternating inputs 16 and 17 belong to one duplicate group. The complete-link groups defined on the twenty-input set are restricted to the first ten inputs for a consistent nested comparison; each well contributes once and search-hit frequency is not a statistical weight.

At a 50 cm^{-1} rotor cutoff and a 0.10 Å duplicate threshold, expanding ten inputs to twenty changes finite-set ensemble G by -1.03 and -1.23 kJ mol^{-1} for the all-same and alternating backbones, respectively (Figure S10). Minimum-E, minimum-G and ensemble-G changes remain within the declared 2 kJ mol^{-1} criterion across all nine rotor/duplicate settings. This establishes batch stability for both empty components within the selected inputs. It does not prove global conformational completeness, remove the saved-search limits or control possible overlap between soft-mode qRRHO entropy and explicit conformer partitioning. The configurations are partitioned separately.

The occupied sets remain incomplete. Each occupancy graph and backbone configuration requires its own assessed sampling before a balanced ensemble cycle can be interpreted. These alternate-model controls cannot replace components of the primary-model PMDI or phthalimide cycles.

The intact-segment decomposition is complete for three sampled pairs using six frozen-fragment single-point calculations (Figure S11). Its partition separates interactions at the complex geometry from deformation relative to relaxed fragments. The interaction energies are -55.25, -44.30 and -50.87 kJ mol^{-1}, with deformation terms $+0.56$, $+0.49$ and $+0.38$ kJ mol^{-1}, respectively. A saved-coordinate audit shows 57–93 optimization steps and heavy-atom input-to-final RMSDs of 0.25–0.70 Å for the complexes, so small deformation sums do not imply negligible structural relaxation. Two

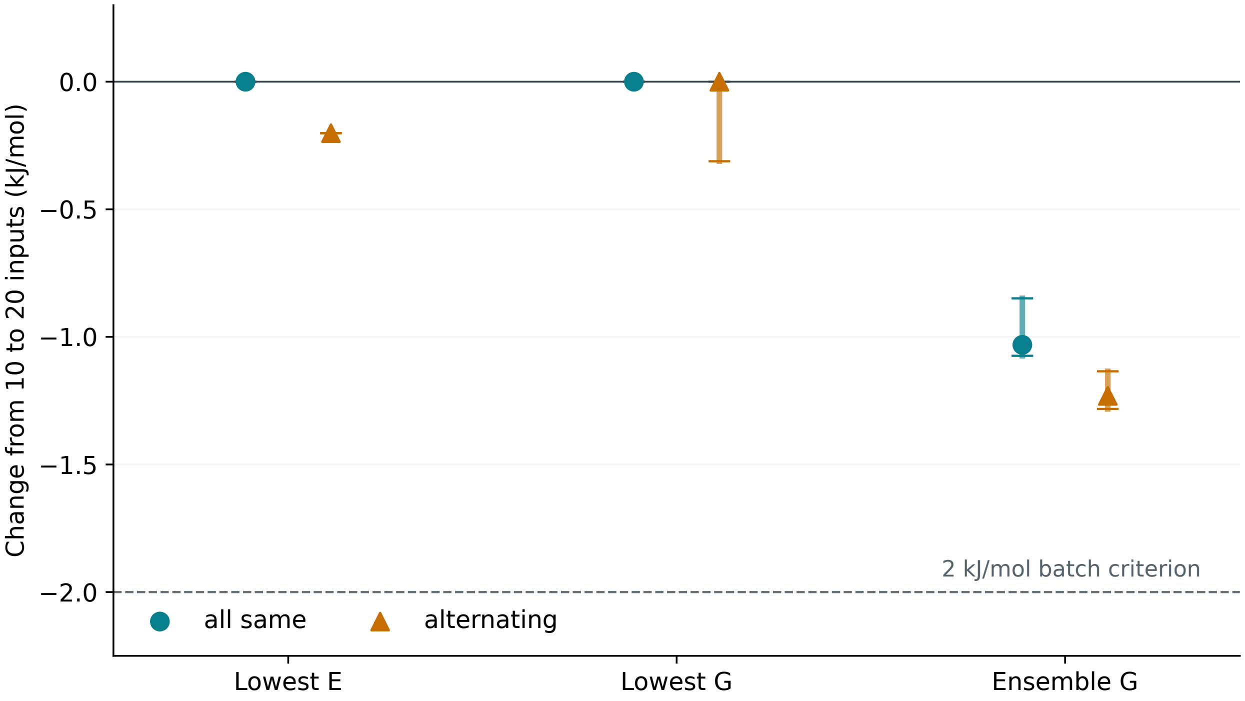


Figure S10: Empty-backbone sampling stability in the separate smooth-surface MeCN model at 298.15 K. Points show changes from the first ten to all twenty inputs at a 50 cm^{-1} rotor cutoff and a 0.10 Å duplicate threshold. Vertical ranges span the explicit rotor/duplicate settings and are not confidence intervals. The dashed line marks the magnitude of the 2 kJ mol^{-1} batch criterion. Each configuration is treated separately; passing empty-component checks does not establish occupied-cycle convergence.

totals also contain cancellation: the parallel phthalimide fragments contribute −0.68 and +1.24 kJ mol^{-1}, and the PMDI fragments contribute +1.00 and −0.62 kJ mol^{-1}. These signed differences refer to one sampled independently optimized monomer at a distinct geometry level; they are not positive-definite strain penalties. Global isolated-monomer minima and Hessian minima are unestablished. The resulting quantities remain electronic, uncorrected for basis-set superposition error and inclusive of backbone and cap contributions. No electron-transfer rate follows from their magnitude.

6 Numerical implementation checks

Diagnostics exposed distinct numerical issues. At a checkpoint of the open PMDI trimer, a legacy solvent cavitation–dispersion–solvent-structure (CDS) force component disagrees with total-energy finite differences by approximately 1.81×10^{-4} hartree bohr^{-1} at the tested coordinate on both integration grids. This discrepancy is not resolved by electronic-grid refinement. Conservative CDS energy derivatives reproduce the tested total-energy differences; their step consistency is checked before use in a continuation. Adaptive refinement of unresolved CDS derivative components recovers the open-trimer optimization. At the returned geometry, independent total-energy derivatives at two step sizes and two tested coordinates agree within 4.2–5.0×10^{-7} hartree bohr^{-1}, below the

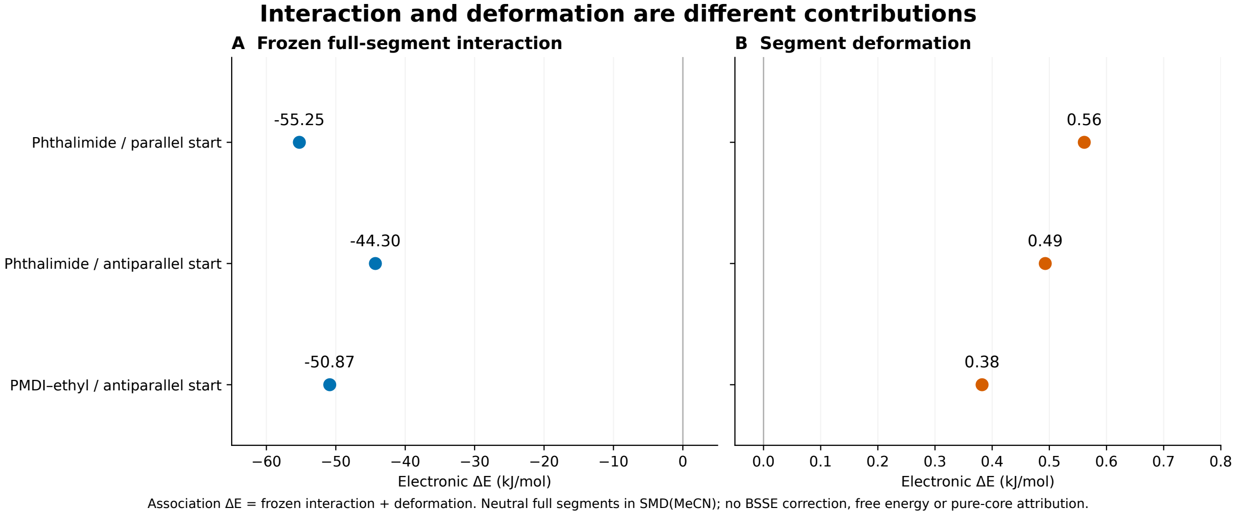


Figure S11: Electronic interaction/deformation partition for three intact segment pairs. Frozen fragments retain the complex geometry without bond cutting or new caps. The partition closes the association-energy cycle but is not a pure pendant-core interaction, a counterpoise-corrected energy or an association free energy.

unchanged 10^{-6} check threshold. This establishes an admitted stationary electronic geometry; no native-Hessian minimum or association free energy is inferred for that original oligomer case.

For the largest spacing Hessian, the original CDS finite-difference step produces a total asymmetry of 2.06×10^{-6}, exceeding the fixed 10^{-6} admission threshold. A separately qualified finer step reduces this to 2.05×10^{-8}. Additive reconstruction replays complete singlet and radical recomputations with maximum matrix differences of 1.33×10^{-11} and 2.83×10^{-9} atomic units, and thermal differences below 1.2×10^{-8} eV. The recovered target then passes the ordinary native gates, with a lowest internal frequency of $+7.23$ cm^{-1}. These are execution-equivalence results, not a relaxation of physical QC or a high-level validation of grafting energetics.

6.1 Solvent-surface differentiability controls

Harmonic thermochemistry requires a locally differentiable energy surface. Surface discretization can introduce artifacts into continuum-solvent geometries and vibrational derivatives; switching/Gaussian constructions address this numerical requirement.[20] At a tested 4.2 Å carbon contact in an excluded backbone structure, the centered curvature of the legacy CDS component increases approximately as the inverse displacement. A separate smooth-surface control instead gives a stable curvature over the tested displacements (Figure S12). This is a component diagnostic, not a full DFT Hessian or a polymer free energy.

The control replaces only the nonelectronic area term with the installed experimental PySCF solvent-accessible-surface formulation at 2,030 points per sphere, retaining the SMD electrostatic surface at 590 points per sphere. The implementation is treated as a distinct model rather than relabeled as the primary calculation. Fixed-geometry component energies, quadrature and rigid-rotation sensitivity are checked separately from derivatives. All Cartesian CDS gradient components of the stored imide precursor, DBU, protonated DBU and chloride-pair structures pass central energy differences at 10^{-4}, 5×10^{-4} and 10^{-3} bohr with maximum errors below 8×10^{-9} hartree bohr^{-1}.

Full quantum controls cover formaldehyde, methyl radical, ammonia and chloromethyl radical. Both surface models are reoptimized independently; full SMD energy differences test all twelve Cartesian gradient components, while two Hessian columns are tested at 0.0005 and 0.001 bohr. The smooth controls pass the unchanged 10^{-6} hartree bohr^{-1} force and 1% Hessian criteria (Table S12). Projected modes and nuclear sums are independently reconstructed. A common imposed symmetry number of one is used solely for these nuclear-kernel comparisons; primary state calculations retain their stated symmetry procedure. These controls establish numerical self-consistency within their neutral molecular scope, not predictive solvation accuracy, polymer ensemble convergence or synthetic eligibility. Primary benchmark and grafting values remain at their original model definition; any alternative-model reaction or occupancy cycle requires all of its components to be calculated consistently.

Table S12: Smooth-surface quantum controls. Force errors are maxima over twelve Cartesian components at both steps; Hessian errors are maxima over the two tested columns, reported as percentages. Frequencies are the lowest projected internal modes in cm^{-1}.

Control	max $\lvert\delta g\rvert$ / hartree bohr^{-1}	max δH / %	ν_{min}
Formaldehyde	5.14×10^{-7}	0.0013	1182.1
Methyl radical	2.19×10^{-7}	0.0149	525.4
Ammonia	1.98×10^{-7}	0.0003	1127.5
Chloromethyl radical	1.32×10^{-7}	0.2793	190.3

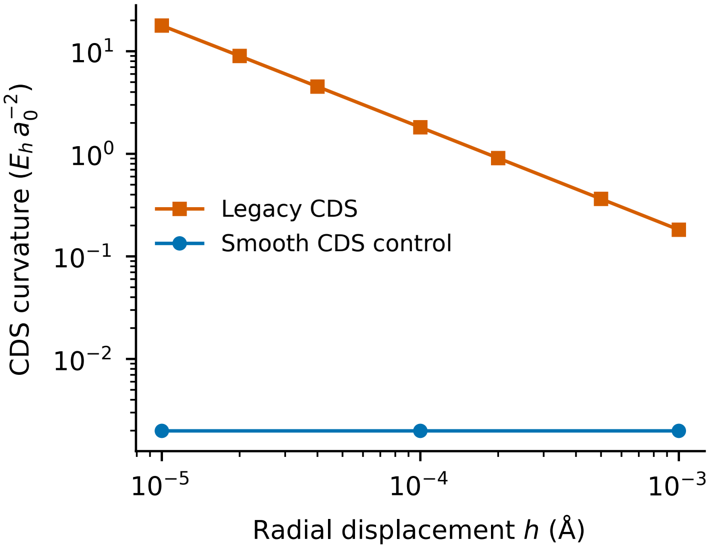


Figure S12: Radial finite-difference curvature of the CDS component at the tested carbon contact. The legacy estimate does not converge as the displacement shrinks, whereas the separate smooth control approaches a plateau. No full electronic Hessian or thermochemical result is inferred from this component scan.

7 Execution and restart controls

A synthetic control packs one million distinct tasks into one thousand chunks in 75.9 s with a measured preparation peak RSS of 67.2 MiB. Eight workers on two hosts execute the synthetic tasks over a 9.58 s span, excluding packing and startup. Independent arithmetic checks find zero discrepancies. This timing characterizes scheduling and storage, not chemical enumeration, conformer search or DFT throughput.

Separately, all 837 molecular descriptor outputs agree with serial definitions. In the shared-filesystem recovery experiment, 32 real design recipes deduplicate to 17 exact molecular tasks. A worker on one host exits abruptly after four checkpointed tasks; owner verification on that host precedes recovery. A second host completes the remaining thirteen tasks without repeating the four checked tasks. All resumed outputs match uninterrupted execution and a separate serial replay. This experiment tests descriptor-stage recovery. A distinct quantum interruption experiment is described below; neither establishes million-molecule quantum throughput. A seeded randomized concurrency experiment executes the same 837 exact designs fifteen times, with three repeats per condition and eight tasks per chunk. All 12,555 molecular comparisons match the independent serial definition. Median launch-to-observed completion times are 4.436, 3.147, 2.680 and 2.497 s for one, two, four and eight CPU workers on one host; eight workers across two hosts require 2.585 s (Figure S13). These times include node preflight, SSH, scorer initialization, execution and at most 0.1 s of observation polling; preparation is separate. All requested workers finish, but only seven of eight execute chunks in one single-host repeat. The diminishing returns of this small library do not determine an optimal concurrency for larger or more expensive stages.

A separate sparse Git checkout and newly installed Python 3.11 CPU environment reproduce the declared 32-recipe / 17-design example with system and user site packages disabled. The project loads from that checkout and scientific dependencies load from the new environment. Serial descriptor replay, exact-route holds and the regenerated conditional-grafting table all match; rendered RGB pixels of the figure are identical. This qualifies a small CPU reproduction using curated scientific data. It does not recompute the underlying quantum energies or qualify the complete CUDA environment.

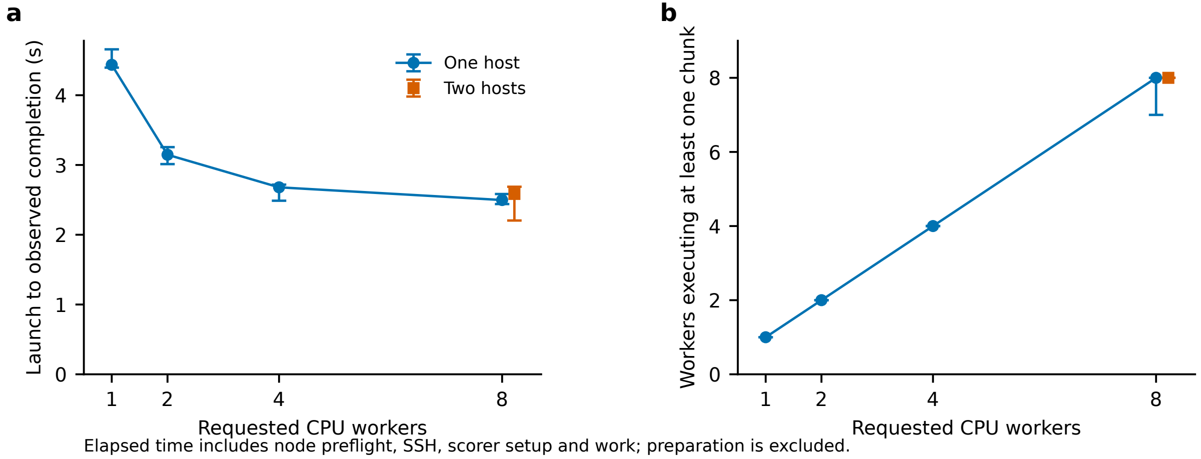
These throughput measurements apply to the measured descriptor and scheduling stages. They neither quantify full quantum-campaign throughput nor establish recovery for arbitrary flexible-polymer calculations.

7.1 Actual quantum interruption and recovery

Two rigid controls, formaldehyde (singlet) and methyl radical (doublet), compare uninterrupted calculations with a fresh process resumed after an intentional exit following three durable gradient evaluations. The exit code is 87; process and start identity confirm termination before resubmission. The saved geometry is unqualified until optimization, active electronic energy, DFT Hessian, projected partition, topology and spin checks pass normally. Acceptance tolerances were 10^{-5} eV for electronic energy, 10^{-4} eV for free energy, 10^{-3} Å aligned RMSD and 0.5 cm^{-1} for internal frequencies. Singlet differences are 4.03×10^{-6} eV, 1.26×10^{-5} eV, 6.87×10^{-6} Å and 0.0685 cm^{-1}; doublet differences are 4.00×10^{-6} eV, 1.11×10^{-6} eV, 3.45×10^{-5} Å and 0.0264 cm^{-1}. Four final partitions were independently reconstructed. This verifies restart equivalence for these two controls, not flexible-polymer sampling or chemical accuracy.

Measured CPU descriptor concurrency: 837 exact designs

Same 1,593 recipes · 8 tasks per chunk · 3 repeats per condition



Elapsed time includes node preflight, SSH, scorer setup and work; preparation is excluded.

Ranges are observed min–max, not confidence intervals. Shared nodes and a small molecular library.

This measures descriptor execution; it does not measure million-molecule chemistry or quantum throughput.

Figure S13: Measured CPU descriptor concurrency for the same 837 exact designs. Three randomized repeats per condition retain every result. Ranges are observed minima and maxima, not confidence intervals. Launch, preflight and scorer overhead are included; preparation is separate. Requested workers and workers executing at least one chunk are distinct. This shared-node, small-library experiment does not measure million-molecule chemistry or quantum throughput.

8 Reproducibility package

A locally prepared reviewer package contains 243 thermochemical state records and 32 electronic-only records, together with versioned code, exact conda artifacts, pip distribution versions and an immutable evidence snapshot. Its 2,498 archived files are hash-verified. After relocation, a standard-library interpreter running in isolation reconstructs all 39 benchmark endpoints, 67 HCl-reference grafting values, 20 base-assisted values, two relative-basicity controls and six finite graph partitions. The auxiliary replay covers five alternative-basis controls, two interruption/recovery comparisons and three electronic fragment-decomposition cycles. A separate CPU reconstruction recovers internal modes and nuclear partitions for 242 non-monatomic states from their stored DFT Hessians; it does not load the xTB library or repeat the original physical minimum checks. Seven changed-data, identity, balance, path and unsupported-promotion tests are rejected.

The package represents the preceding numerical evidence epoch, `2026-10-23_all_axes_checked`; subsequent records are frozen separately in the present manuscript snapshot. A fourteen-state PMDI addendum includes the two singly occupied chains, their matched empty chain, the doubly grafted chain and molecular/reagent references. After archive extraction and relocation, isolated standard Python reconstructs fourteen balanced reaction comparisons and the complete neighboring-graft cycle at 25, 50 and 100 cm^{-1}; both attachment orders close. A fresh environment containing only hash-locked NumPy 2.2.6 independently reconstructs thirteen DFT-Hessian mode sets and their nuclear sums at all three cutoffs, plus monatomic chloride translation. Original project reads are blocked in both modes, and no quantum program or xTB library is loaded. Fourteen integration checks include rejection of changed or borrowed Hessians, altered masses, symmetry numbers and frequencies, as well as identity, arithmetic, scope and path failures. These cached-data checks do not repeat the original SCF, stationarity, spin or displaced-gradient validation.

An unpublished repository draft stages the earlier package; downloaded assets match the recorded checksums. The PMDI and control addenda remain local reviewer artifacts. Package replay establishes numerical traceability within the declared scopes, rather than experimental accuracy or broader conformer convergence. No public DOI or deposition is claimed. General GPU-platform portability, remaining execution-diagnostic records and final deposition metadata remain open; the separately tested deployment profile is described below.

A separate TEMPO addendum contains the two local-graft states and the two Fc reference states. After relocation, standard-library arithmetic reproduces the three rotor-cutoff potentials, and locked NumPy reconstructs the four Hessian mode sets and nuclear terms with original-project Python opens blocked. These records overlap the earlier packages and are not counted as additional unique Fc states. The replay does not repeat SCF, stationarity or spin checks.

A complementary numerical-control addendum contains eight small-molecule geometry/Hessian records for four species under two surface models, together with six fixed-geometry Fc electronic records. Its 392 stored SCF evaluations comprise eight baselines and 384 displacements. A hash-locked, freshly installed NumPy CPU environment reconstructs the stored force differences, sampled Hessian columns, internal modes and nuclear sums after archive extraction and relocation, with original project reads blocked. A standard-library mode reconstructs the Fc and component arithmetic. These cached-data checks neither rerun the physical calculations nor qualify a new polymer ensemble; identities overlap with the other packages and record counts must not be summed as unique molecules.

8.1 Source-bound reconstruction of retained polymer states

The sixty retained smooth-surface polymer records were read after an explicit checkout-prefix relocation, without rewriting their original paths or hashes. A relative source path was accepted only with a declared original working directory. Missing archived files could not fall back to files in the current checkout. The replay verified 12,386 source/input bindings and forced a fresh content-hash pass over all bindings before publishing a completed report.

For each state, the replay checked the exact graph, neutral-singlet identity, solvent-model definition and equality of the thermochemical and scored electronic geometries. It reconstructed the stored sampled Hessian columns from the displaced gradients at 0.0005 and 0.001 bohr, including the protocol coordinates and the two largest-force coordinates. The rigid-space projection, internal frequencies, symmetry number, three rotor-cutoff nuclear partitions and final electronic/nuclear/spin sums were then reconstructed. Retained stationarity, exclusive-residency and resource records were checked separately. Fifteen further completed states passed the same physical reconstruction after explicit output-path relocation. The forty empty-backbone inputs analyzed above are fully included; reconstructed occupied states remain individual controls until their sets are complete. The following numerical maxima refer to the original sixty-state replay. The largest sampled-column relative error was 3.92×10^{-5}; the maximum differences from stored internal frequencies and nuclear free energies were 6.97×10^{-11} cm^{-1} and 1.50×10^{-13} eV. Seventeen software tests covered the archive binding and derivative reconstruction, including rejection of changed bytes, incorrect displacements, nonfinite force/spin values and contradictory backbone labels. These checks establish reproduction of the saved physical evidence; they do not repeat SCF, optimization or Hessian execution, qualify the current GPU runtime, or supply missing conformers.

A portable numerical-input archive contains the geometries, Cartesian Hessians, gradients, displaced-gradient records, electronic parents and three-cutoff partitions for all 75 reconstructed expanded-polymer states. After extraction, a newly installed, hash-locked NumPy 2.2.6 environment reproduces the stored sampled derivative checks, internal-mode projections and nuclear free-energy terms with access to the original checkout, archive, calculation directories and prepared cache blocked. The unchanged original reviews retain the hashes of historical executable binaries and large response scratch tensors that are excluded from this numerical-input archive. The replay reconstructs saved numerical evidence; it does not repeat SCF, live execution checks or the complete historical source closure. Rehashed but inconsistent Hessians, gradients, masses, solvent identities and nonfinite quantities are rejected.

8.2 Fresh quantum-environment reproduction

A separate CUDA 12.8 compatibility profile retains the recorded molecular-method packages and xTB build while changing fourteen CUDA runtime distributions to satisfy the exact requirements of Torch 2.8.0+cu128. The historical runtime inventory remains separate. All 235 external distributions were installed into a fresh prefix, the project was installed from an archived source checkout, and the dependency check passed. Retrieved package artifacts carry SHA-256 identities; the conda artifact lock and all certificate backing files are retained. The fresh xTB library is byte-identical to the qualified thermodynamic build.

Formaldehyde (singlet) and methyl radical (doublet) were independently optimized and evaluated with the active electronic-energy, native Hessian and projected-partition procedures in the fresh environment. Both passed the normal physical checks and the predeclared original-versus-fresh comparison bounds: 10^{-5} eV for electronic energy, 10^{-4} eV for free energy, 10^{-3} Å for aligned RMSD and 0.5 cm^{-1} for internal frequencies. A separate replay reconstructed both saved Hessian mode sets and all three rotor-cutoff partitions. Package imports, shared-library locations and the thermodynamic helper interpreter were checked against the fresh prefix; the original environment was not modified. These controls establish reproduction on the tested Linux/CUDA host. They do not establish unrestricted device coverage, eliminate external cluster-reservation configuration or extend polymer conformer validation. The deployment evidence is separate from the earlier cached reviewer-package epoch.

8.3 Relocation of saved native-Hessian partitions

Saved formaldehyde and methyl-radical records were copied with their native Hessians and certificate backing files into a separate checkout. An explicit original-prefix declaration maps the stored certificate paths onto local files only when their hashes and all other certificate metadata agree. The original physical checks remain in force, and the native records are unchanged. With the separately staged, hash-qualified xTB prefix, both projected partitions were recomputed through the CPU entry point. Internal frequencies, symmetry numbers, spin terms and nuclear free energies at all three rotor cutoffs matched their original values exactly. Python reads of the original checkout were blocked; a file-system trace across the process tree recorded no access to files inside it. Ten altered-input controls were rejected, including changed stationarity, gradient, spin, derivative and certificate evidence. These checks establish the tested relocation of saved-Hessian postprocessing; they do not repeat electronic-structure calculations or requalify native GPU Hessian execution.

8.4 Molecular warm-start execution profiles

Gas-phase molecular warm starts retain their recorded engine identities, with the standard engine selected on V100. The tested UMA Nitro profile is not accepted as a production calculator. In a fixed-geometry repeatability control on the 42-atom neutral PMDI-ethyl design using identical checkpoint weights, the initial-to-final force difference was 7.57×10^{-4} eV Å^{-1} for Nitro, above the predeclared 10^{-4} eV Å^{-1} bound; the standard profile gave 1.65×10^{-6} eV Å^{-1}. This single-case failure does not identify its implementation cause or establish full-panel parity.[21] Composition, charge and spin define each calculator state; a state change creates a new fixed-state predictor and invalidates cached ASE results. Checkpoint weights, atomic references, loader identity and actual execution mode identify the model profile. Current-model warm starts occupy a separate cache, while archived calculations retain their original model and input identities. Solution energies and nuclear partitions follow the DFT/SMD procedures described above. Numerical comparisons of starting-geometry engines do not establish solution thermochemistry or full polymer conformer convergence.

9 Evidence terminology

Table S13: Terms used in evidence files and their interpretation.

Term	Meaning
Exact design	A specified precursor/product graph and mapped attachment bond; generating recipes can deduplicate to the same design.
Checked or qualified	The checks required for the named field passed; this is not experimental validation.
Native free energy	Active electronic energy plus the projected geometry-level DFT nuclear partition and declared spin partition.
Frozen snapshot	An immutable, hash-verified copy of specified evidence; newer live results require a new snapshot.
Proceed / hold / exclude	Route decisions within the stated evidence scope; hold means required exact-route evidence is incomplete.
Endpoint	One sourced redox-potential comparison, with exact molecular identity, state pair and reference convention.
Pose	One conformational starting geometry; the final optimized contact class is checked separately.

References

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